IoE and AI driven systems
LAB OUTPUTS

NAME:**FIONA PRISCILLA MIT S.** REG NO:**1912260145**

EXPERIMENT NO:4

Simulation and Comparative Performance Evaluation of Zigbee, 6LoWPAN and NB-IoT Communication Protocols for IoE Applications using NS-3

PROGRAM CODE:

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

np.random.seed(24)

protocols=["Zigbee","6LoWPAN","NB-IoT"]
simulation_time=120
packets_sent=1200
packet_size=512

parameters={
   "Zigbee":{"pdr":92,"delay":38,"energy":15,"throughput":41},
   "6LoWPAN":{"pdr":89,"delay":62,"energy":21,"throughput":34},
   "NB-IoT":{"pdr":97,"delay":72,"energy":29,"throughput":45}
}

results=[]

for protocol in protocols:
   p=parameters[protocol]
   pdr=np.clip(p["pdr"]+np.random.normal(0,1.0),80,100)
   delay=max(1,p["delay"]+np.random.normal(0,2.5))
   energy=max(1,p["energy"]+np.random.normal(0,0.8))
   throughput=max(1,p["throughput"]+np.random.normal(0,1.2))
   packets_received=round(packets_sent*pdr/100)

results.append([protocol,packets_sent,packets_received,round(pdr,2),round(throughput,2),rou
nd(delay,2),round(energy,2)])

df=pd.DataFrame(results,columns=["Protocol","Packets Sent","Packets Received","Packet
Delivery Ratio (%)","Throughput (kbps)","Average Delay (ms)","Energy Consumption (J)"])

print("="*75)
```

```
print("IoE COMMUNICATION PROTOCOL PERFORMANCE ANALYSIS")
print("="*75)
print(df.to_string(index=False))

best_pdr=df.loc[df["Packet Delivery Ratio (%)"].idxmax(),"Protocol"]
best_throughput=df.loc[df["Throughput (kbps)"].idxmax(),"Protocol"]
lowest_delay=df.loc[df["Average Delay (ms)"].idxmin(),"Protocol"]
lowest_energy=df.loc[df["Energy Consumption (J)"].idxmin(),"Protocol"]

print("\nPERFORMANCE SUMMARY")
print("-"*45)
print("Best PDR        :",best_pdr)
print("Best Throughput :",best_throughput)
print("Lowest Delay    :",lowest_delay)
print("Lowest Energy   :",lowest_energy)

metrics=[
   ("Throughput (kbps)","Throughput Comparison","Throughput (kbps)"),
   ("Average Delay (ms)","Average Delay Comparison","Delay (ms)"),
   ("Packet Delivery Ratio (%)","Packet Delivery Ratio Comparison","PDR (%)"),
   ("Energy Consumption (J)","Energy Consumption Comparison","Energy (J)")
]

for column,title,ylabel in metrics:
   plt.figure(figsize=(8,5))
   plt.bar(df["Protocol"],df[column])
   plt.title(title)
   plt.xlabel("Protocol")
   plt.ylabel(ylabel)
   plt.grid(axis="y",alpha=0.3)
   if column=="Packet Delivery Ratio (%)":
      plt.ylim(0,100)
   plt.tight_layout()
   plt.show()

x=np.arange(len(protocols))
width=0.2

fig,ax=plt.subplots(figsize=(10,6))
ax.bar(x-1.5*width,df["Packet Delivery Ratio (%)"],width,label="PDR")
ax.bar(x-0.5*width,df["Throughput (kbps)"],width,label="Throughput")
ax.bar(x+0.5*width,df["Average Delay (ms)"],width,label="Delay")
ax.bar(x+1.5*width,df["Energy Consumption (J)"],width,label="Energy")
ax.set_xticks(x)
```

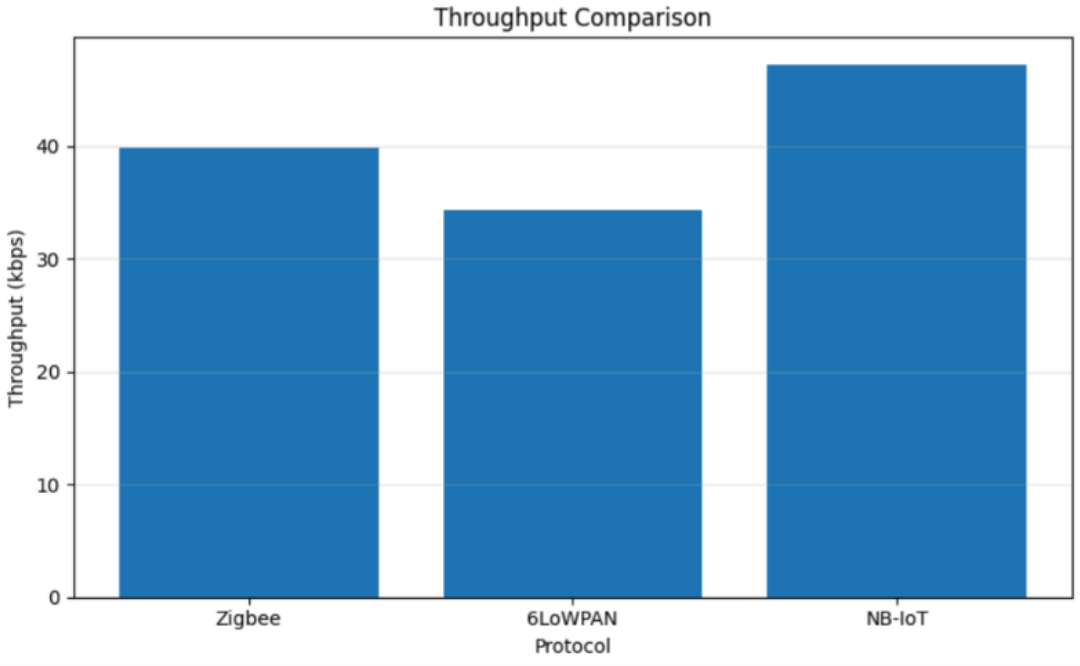
```
ax.set_xticklabels(protocols)
ax.set_xlabel("Communication Protocol")
ax.set_ylabel("Performance Value")
ax.set_title("Comparative Performance of IoE Protocols")
ax.legend()
ax.grid(axis="y",alpha=0.3)
plt.tight_layout()
plt.show()

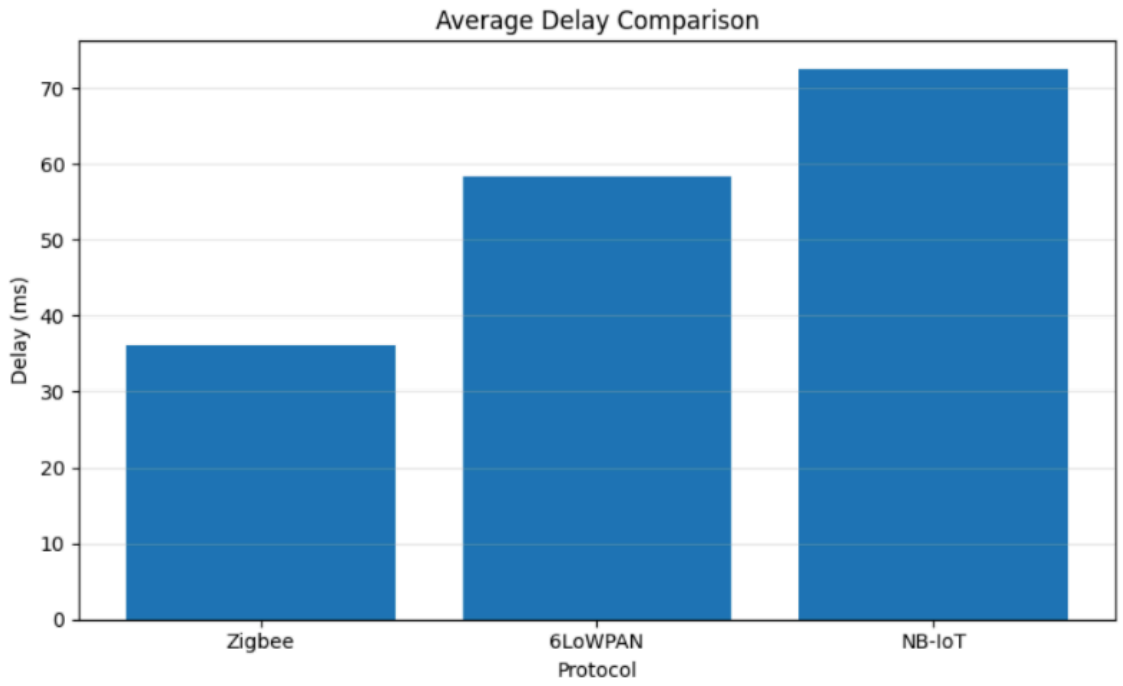
print("\nCONCLUSION")
print("-"*45)
print("Zigbee provides low delay and low energy consumption.")
print("6LoWPAN provides IPv6 connectivity with moderate performance.")
print("NB-IoT provides high PDR and throughput for long-range IoT applications.")
```

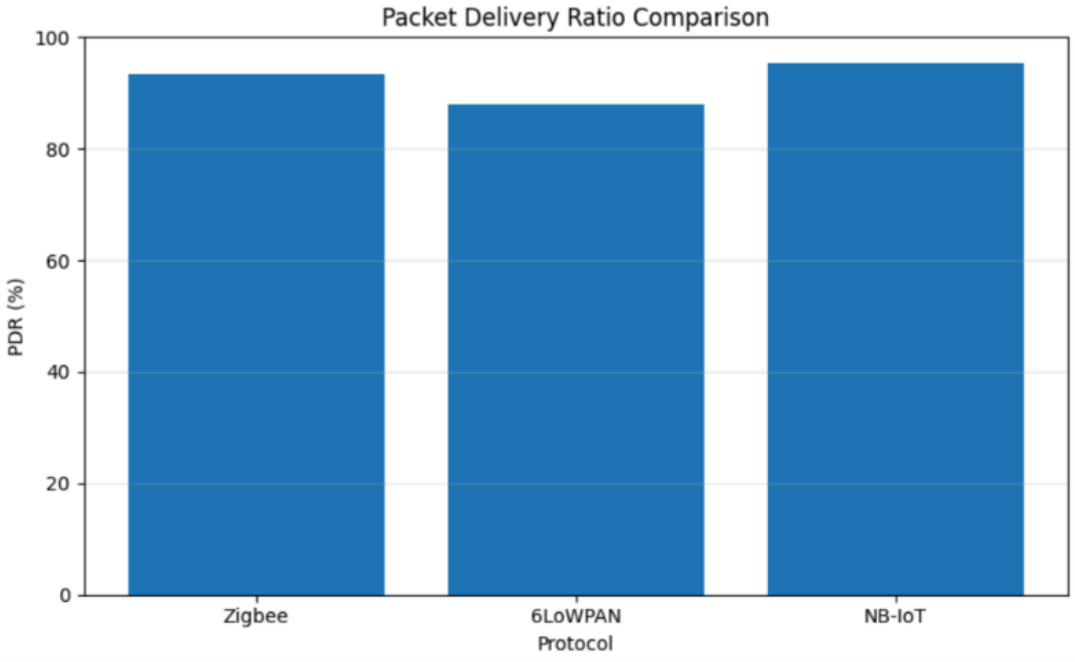
OUTPUTS:

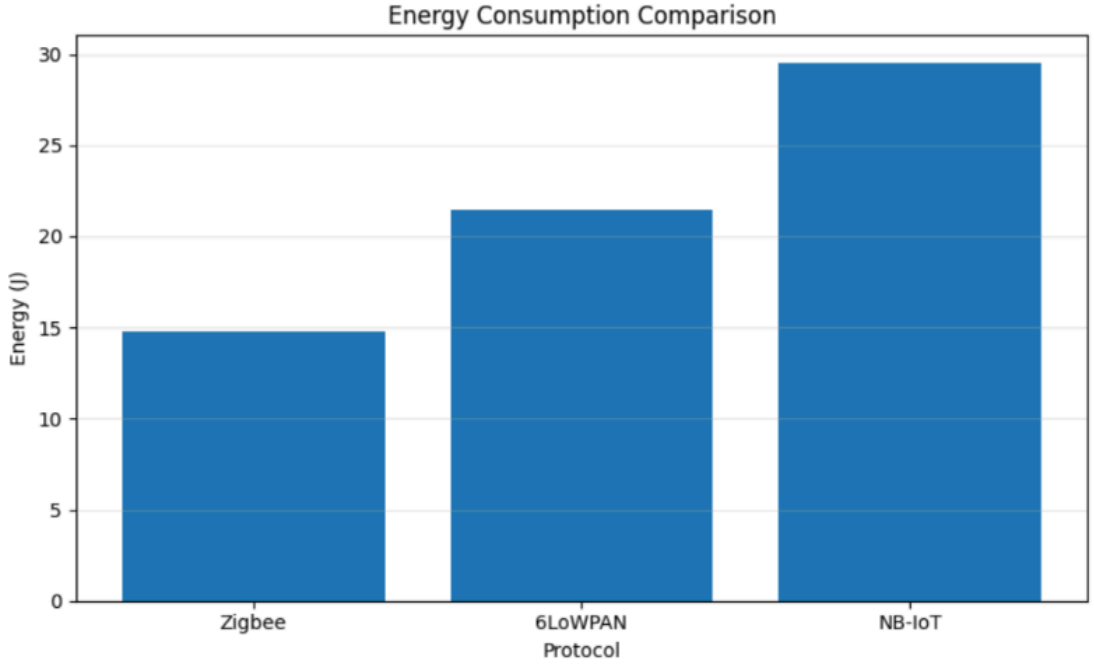
```
======================================================================
IoE COMMUNICATION PROTOCOL PERFORMANCE ANALYSIS
======================================================================
Protocol  Packets Sent  Packets Received  Packet Delivery Ratio (%)  Throughput (kbps)  Average Delay (ms)  Energy Consumption (J)
  Zigbee          1200              1120                     93.33              39.81               36.07                   14.75
 6LoWPAN          1200              1055                     87.93              34.35               58.40                   21.45
  NB-IoT          1200              1144                     95.37              47.27               72.55                   29.54

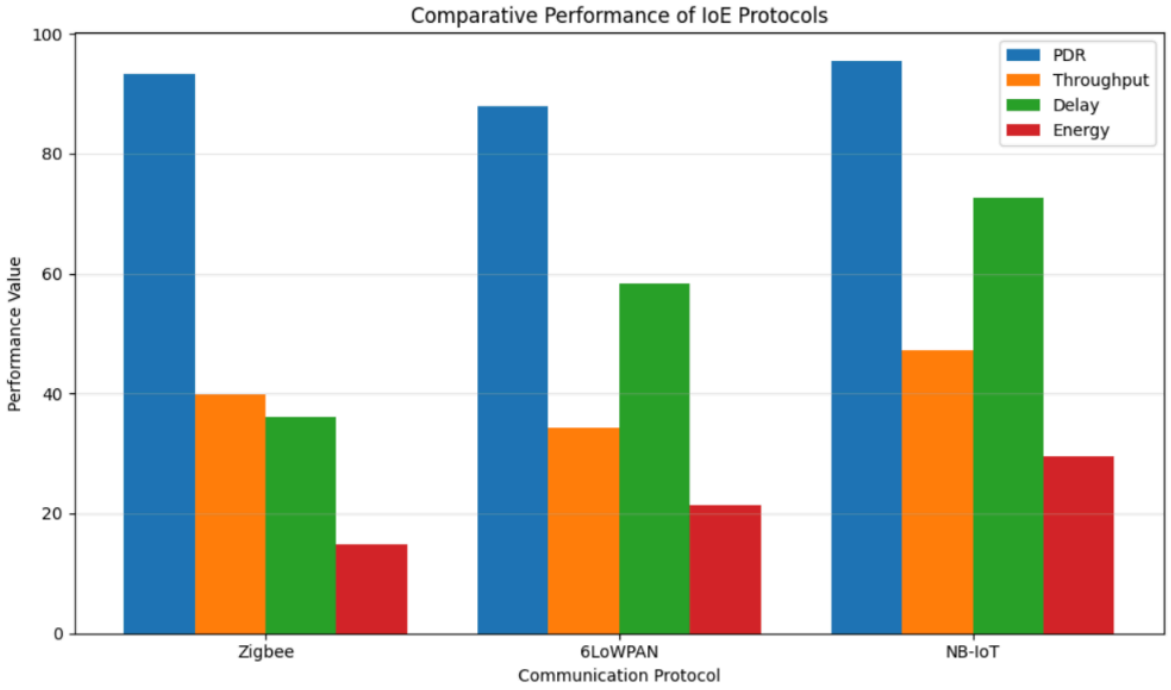
PERFORMANCE SUMMARY
--------------------------------------------
Best PDR        : NB-IoT
Best Throughput : NB-IoT
Lowest Delay    : Zigbee
Lowest Energy   : Zigbee
```




Average Delay Comparison
Delay (ms)
0
10
20
30
40
50
60
70
Zigbee
6LoWPAN
NB-IoT
Protocol


Packet Delivery Ratio Comparison
PDR (%)
0
20
40
60
80
100
Zigbee
6LoWPAN
NB-IoT
Protocol






```
CONCLUSION
-----------------------------------------------
Zigbee provides low delay and low energy consumption.
6LoWPAN provides IPv6 connectivity with moderate performance.
NB-IoT provides high PDR and throughput for long-range IoT applications.
```